Research Summary Sheet - Article 4

Complete the following form for each artifact that validates your problem statement.

Source (APA format and include a link to the source)

Li, Z., & Bingham, L. (2022). *NASA white paper -terramechanics for LTV modeling and simulation*.

https://ntrs.nasa.gov/api/citations/20220010732/downloads/Terramechanics_white_paper.pdf

Artifact Summary

This article focuses on how the rover wheels interact with the lunar soil, affecting the vehicle's movement. It explains models of rolling resistance, soil compression, and traction, to name a few. The research also mentions various conditions where a rover became stuck in the soil. This connects to my HUNCH project since the rover I'm designing needs to have proper wheels and mobility in regards to the cargo it is carrying.

Artifact Critique

This article is relevant because it explains the mechanical forces acting on a lunar cargo robot. It will be useful for HUNCH since it provides models to predict rover performance in lunar terrain. This article only focuses on mobility, which is one part of the robot we need to build. It focuses primarily on simulations rather than prototype testing.

Reliability

Both of the authors are NASA Johnson Space Center researchers specializing in engineering, specifically lunar vehicle mobility. The work was produced for NASA's Lunar Terrain Vehicle research and published through NASA NTRS.